

Early Stopping in Deep Learning

Stop training when the model stops improving on a validation set.

1. What is Early Stopping?

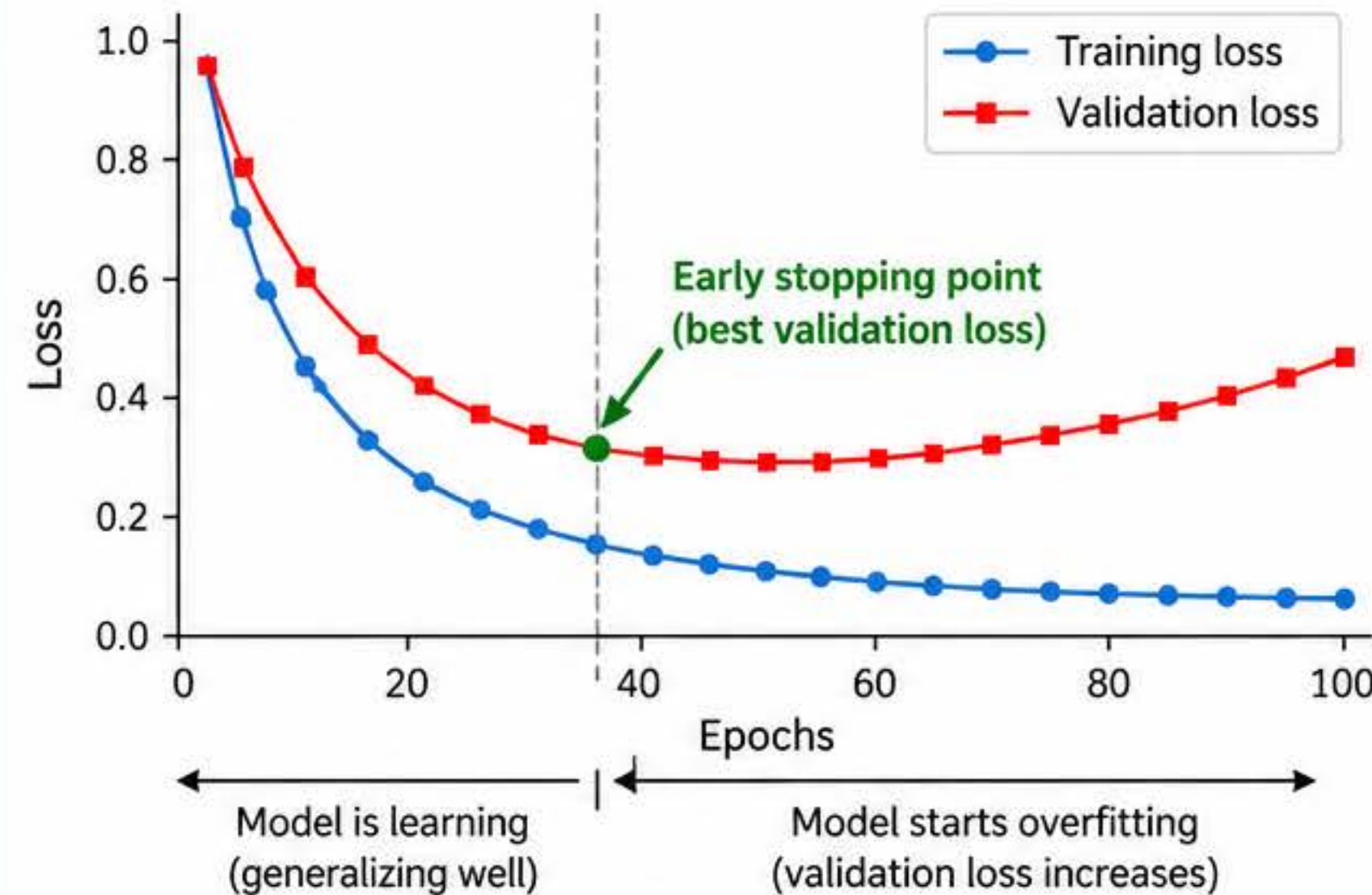
Early stopping is a regularization technique that stops training when the model's performance on a **validation set** stops improving.

Idea: Train the model only as long as it generalizes well, and stop before it starts overfitting.

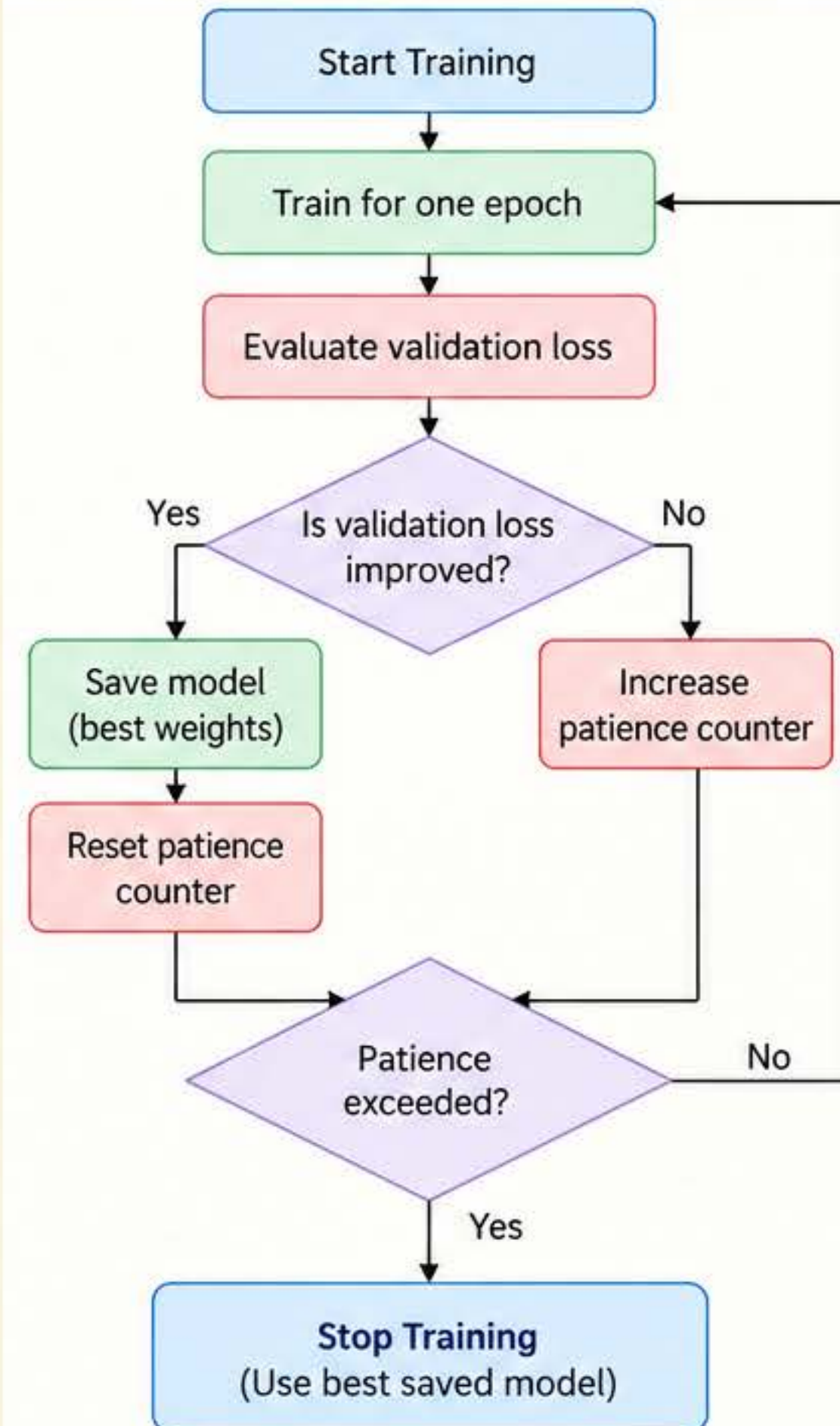
2. Intuition

As training progresses, the model first learns useful patterns, but eventually begins to overfit the training data. Early stopping selects the model at the point of best validation performance.

3. Training and Validation Loss



4. Early Stopping Algorithm



5. Benefits

- 1 Prevents overfitting**
Stops training before the model starts memorizing the training data.
- 2 Improves generalization**
Usually leads to better performance on unseen data.
- 3 Saves computation**
Avoids unnecessary training epochs.
- 4 Simple to implement**
Easy to use with a validation set and a patience value.

6. Important Considerations

- 1 Use a separate validation set**
(Do not use the test set for early stopping).
- 2 Choose a patience value**
(Number of epochs to wait for improvement, e.g., 5–20).
- 3 Monitor an appropriate metric**
(e.g., validation loss, accuracy, F1-score).
- 4 Use the best saved model**
(Weights at the lowest validation loss).

Early stopping is a simple yet effective way to improve generalization by stopping training when the model stops improving on the validation set.

Early Stopping is Equivalent to L2 Regularization

For linear models (and approximately for neural networks), early stopping leads to a solution similar to L2 regularization.

1. Learning Problem

Consider a linear model trained with squared error loss:

$$L(\mathbf{w}) = \frac{1}{N} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2$$

$\mathbf{X}$: input data matrix ($N \times d$)
 $\mathbf{y}$: target vector ($N \times 1$)
 $\mathbf{w}$: model parameters ($d \times 1$)
 N : number of training examples

2. Gradient Descent Solution (No Regularization)

Starting from $\mathbf{w}_0 = 0$, the solution after t iterations of gradient descent with learning rate η is

$$\mathbf{w}_t = (\mathbf{I} - \eta \mathbf{X}^T \mathbf{X})^t \mathbf{X}^T \mathbf{y}$$

Using eigen-decomposition $\mathbf{X}^T \mathbf{X} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T$, this can be written as a shrinkage of each component:

$$\mathbf{w}_t = \mathbf{U} (\mathbf{I} - \eta \mathbf{\Lambda})^t \mathbf{U}^T \mathbf{X}^T \mathbf{y}$$

Each component along an eigenvector u_i is scaled by $(1 - \eta \lambda_i)^t$, which is less than 1.

3. L2 Regularization Solution

The solution with L2 regularization (ridge regression) is

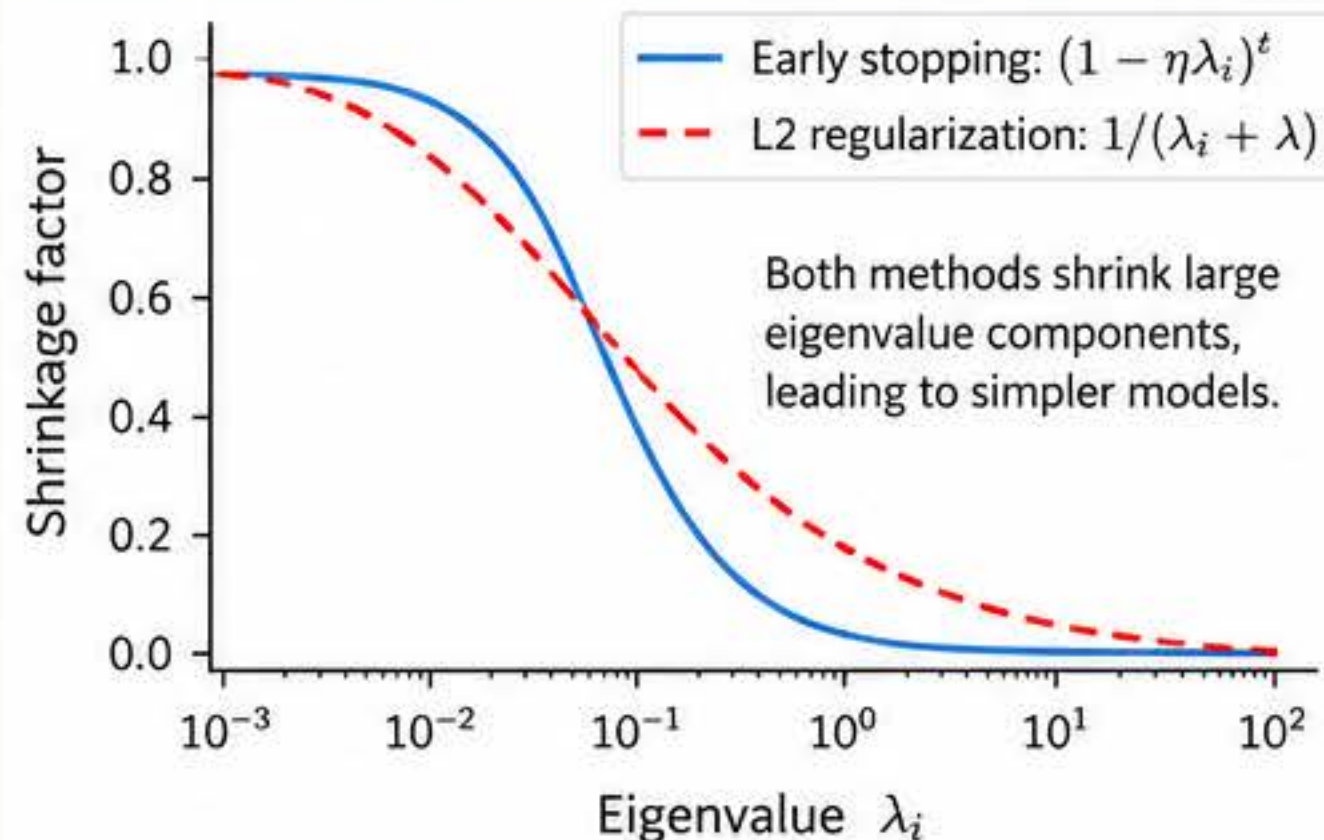
$$\mathbf{w}_{L2} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$$

Using the same eigen-decomposition $\mathbf{X}^T \mathbf{X} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T$,

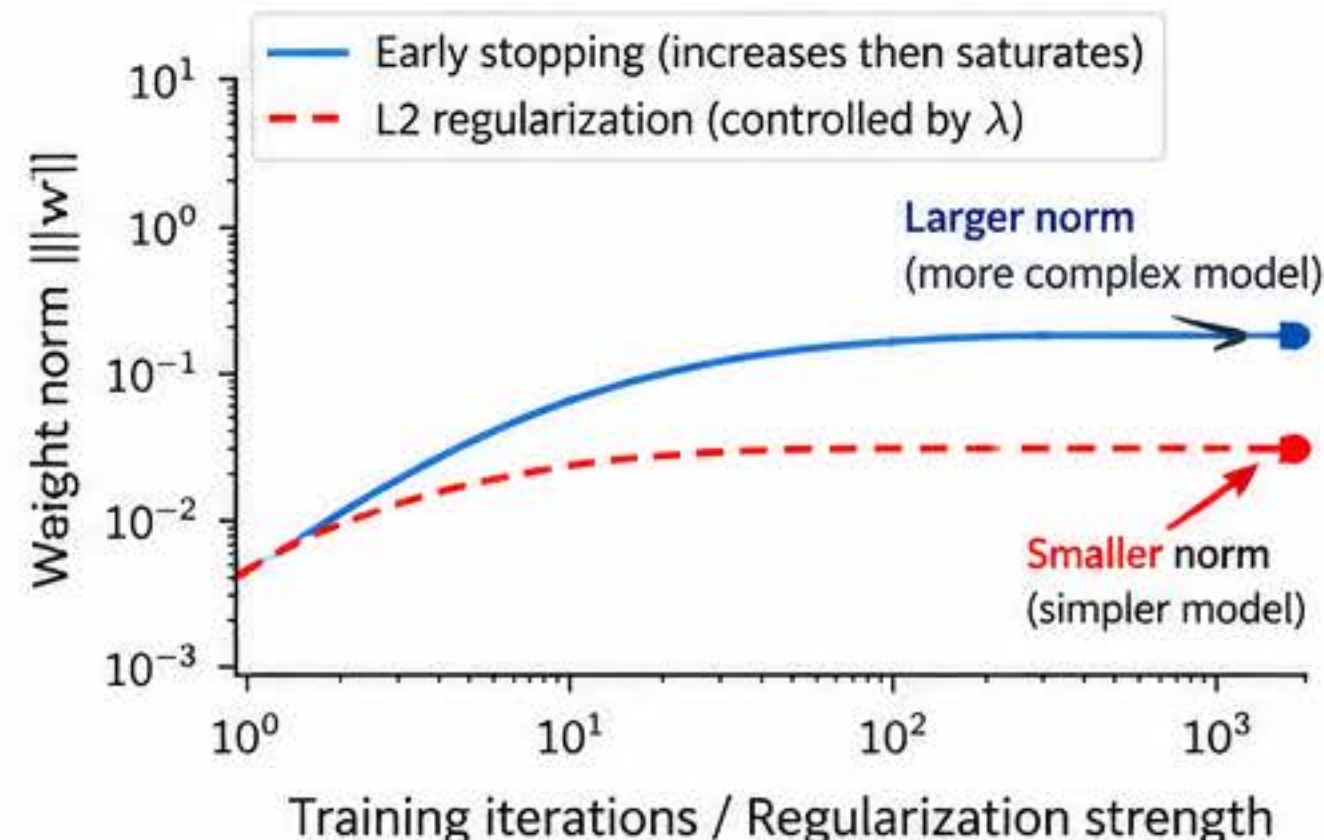
$$\mathbf{w}_{L2} = \mathbf{U} (\mathbf{\Lambda} + \lambda \mathbf{I})^{-1} \mathbf{U}^T \mathbf{X}^T \mathbf{y}$$

Each component along u_i is scaled by $\frac{1}{\lambda_i + \lambda}$ which shrinks large components (high λ_i) more.

4. Comparison of Shrinkage Factors



5. Effect on Model Complexity



6. Intuition and Takeaway

- Both methods shrink weights**
They reduce the magnitude of parameters, especially in directions with large eigenvalues.
- Early stopping = implicit regularization**
Stopping early limits how much each component grows, similar to adding an L2 penalty.
- Equivalent for linear models**
For linear regression, early stopping and L2 regularization produce similar solutions.
- Approximately true for neural networks**
In deep learning, early stopping acts as an implicit form of L2 regularization, though not exactly equivalent.

Early stopping controls the complexity of the model by limiting the growth of weights, which is equivalent to L2 regularization (especially for linear models).